

Visual Complexity of Point Set Mappings

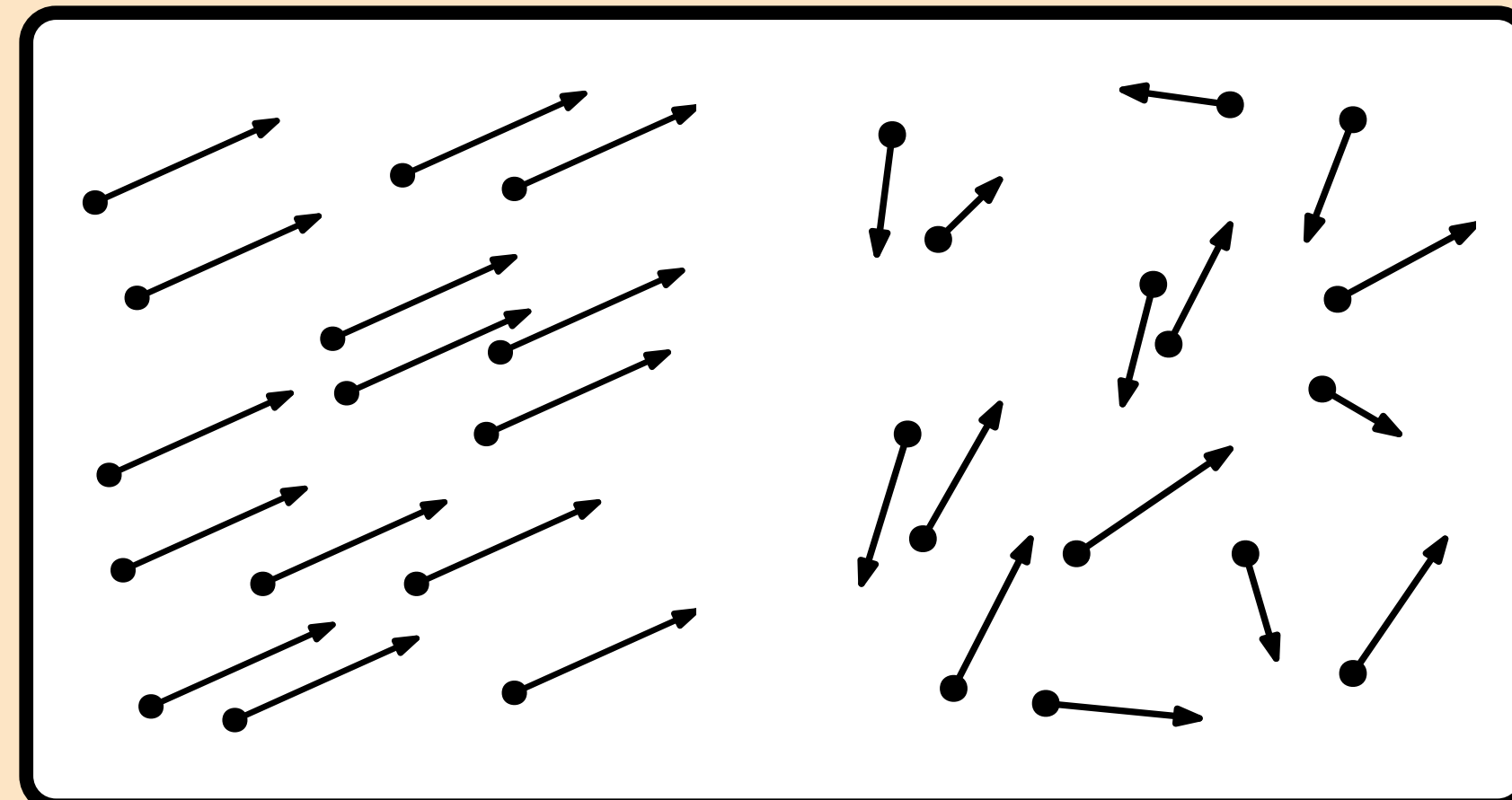


Problem

The position of visual data elements may change in time-varying contexts. This can be visualized using animated transitions or stepwise animations.

Existing **visual complexity measures** of such transitions consider point-set similarities like the sum of point-wise distances.

Such measures **do not** accurately capture the perceptual simplicity of **structured motions**.



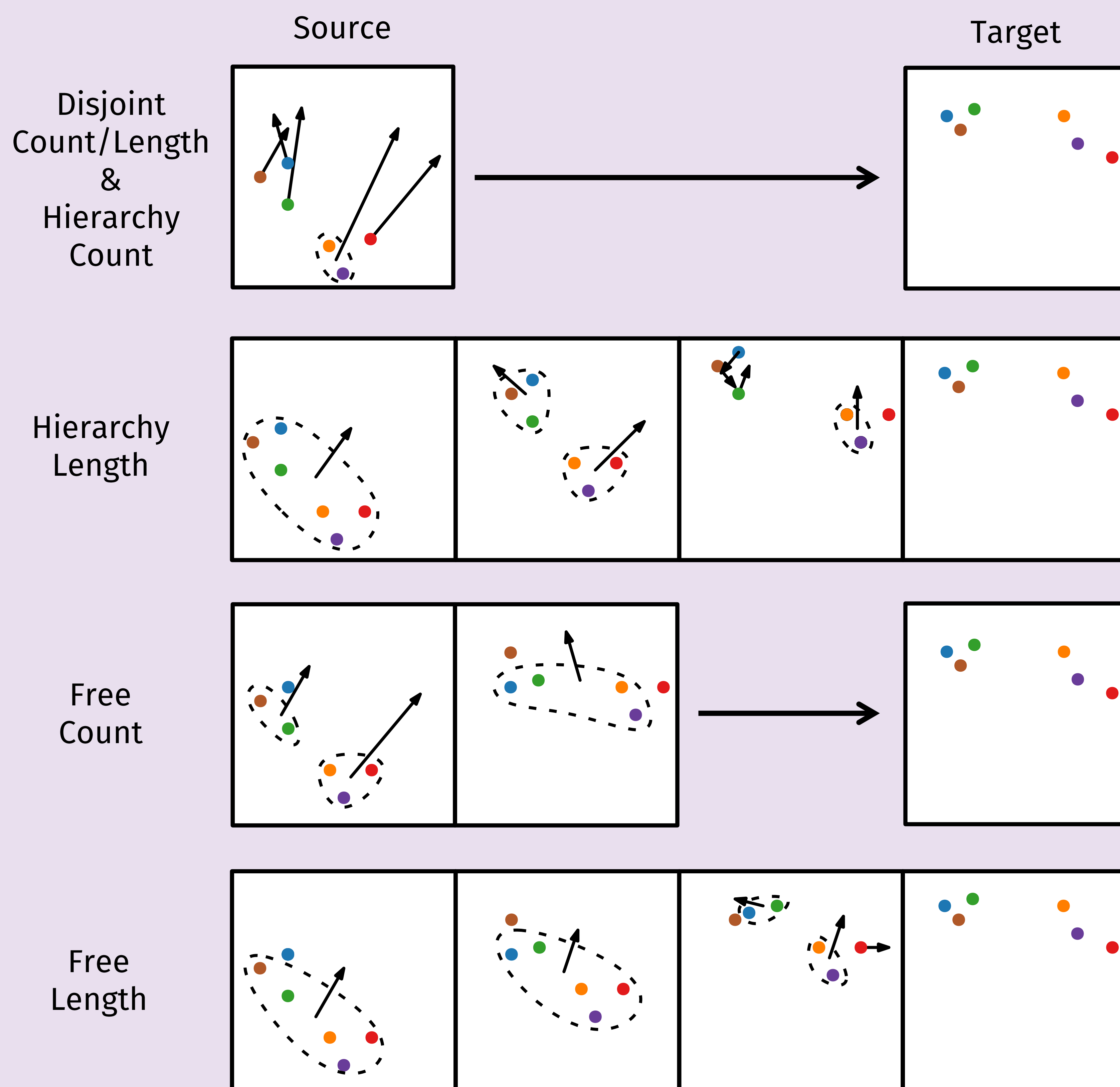
Grouped motion is more perceptually coherent (Gestalt)

Objectives

1
Quantify the **visual complexity** of transitions between point sets in terms of **group translations**.

2
Study the **algorithmic complexity** of the resulting problems.

Framework



Visual Complexity

Translating a group of points counts only once. In doing so, we have two choices:

Permitted groups

- Given
- Disjoint
- Hierarchy
- Free

Group cost

- Count
- Length

Combining these choices permits various ways of modeling visual complexity.

Algorithmic Complexity

Given two point sets, the permitted groups, and the group cost, we minimize the total cost.

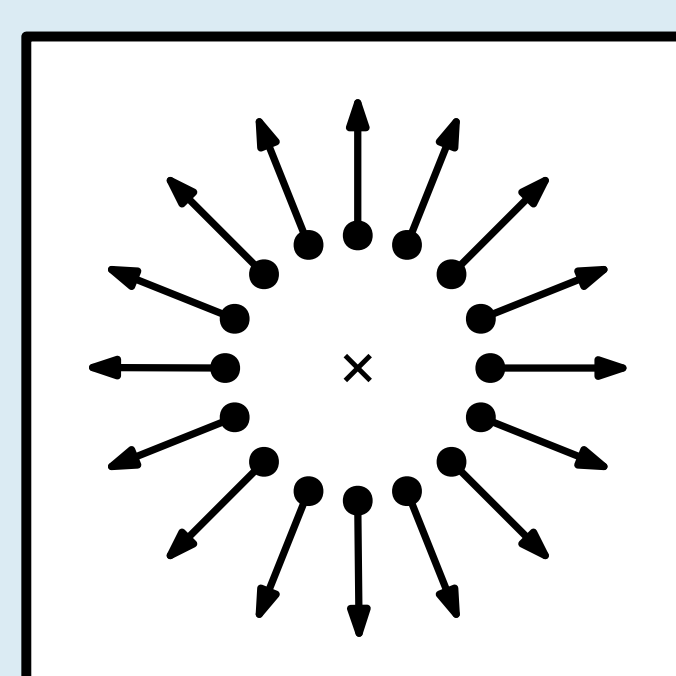
Constraint	d	Count	Length
Given	$d = 1$	n/a	$O(n)^*$
	$d \geq 1$	n/a	Polynomial
Disjoint	$d \geq 1$	$O(dn \log n)$	$O(dn \log n)$
Hierarchy	$d = 1$	$O(n \log n)$	$O(n \log n)$
	$d \geq 2$	$O(dn \log n)$	NP-Hard
Free	$d = 1$	NP-Hard*	$O(n \log n)$
	$d \geq 2$	NP-Hard*	1.307-approx

Extensions

The Framework can be extended in several ways to more accurately capture visual complexity:

Stages

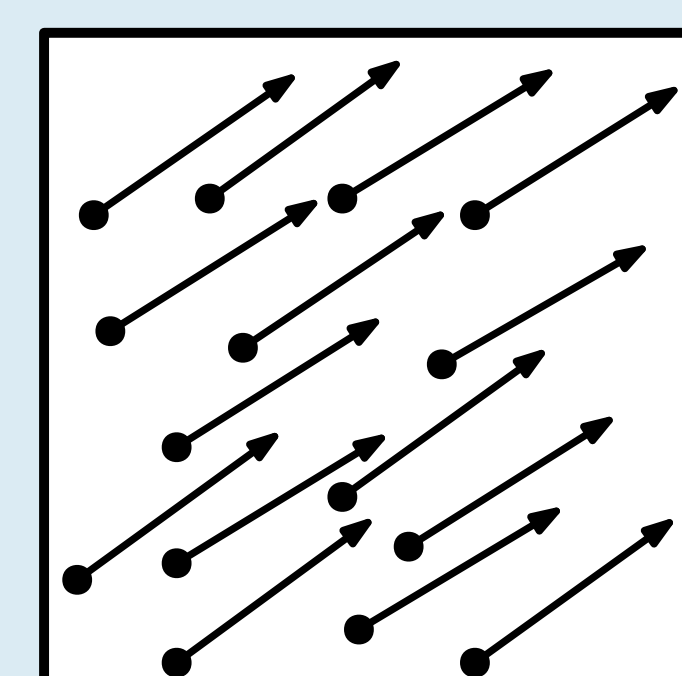
Each square in the framework's transitions represents a "stage". The number of stages can be minimized, or a maximum number of stages can be imposed.



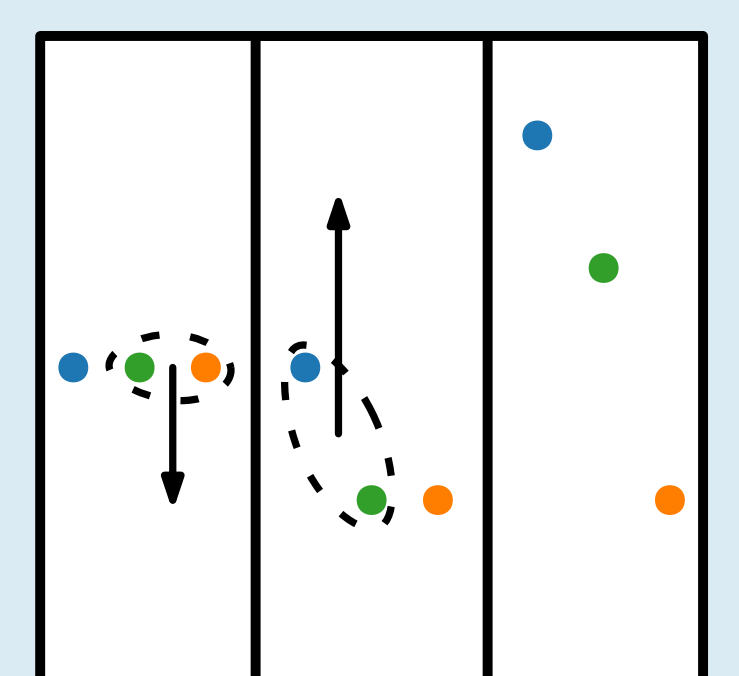
Scaling



Groups should be spatially coherent



Small deviations may be grouped



Avoiding large detours